Statistics and Probability Notes

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1. Descriptive Statistics

Definition: Descriptive statistics is the branch of statistics concerned with organizing, summarizing, and presenting data in a meaningful way.

It describes the main characteristics of a dataset without making conclusions about a larger population.

- Mean
- Median
- Mode
- Range
- Variance
- Standard deviation
- Quartiles
- Correlation

2. Mean

Definition: The mean is a measure of central tendency calculated by adding all observations in a dataset and dividing the total by the number of observations.

It is commonly called the average.

Example: Data: 60, 70, 70, 80, 90. Sum = 370 and n = 5, so mean = 370 / 5 = 74.

Formula: Mean = Sum of all values / Number of observations

3. Median

Definition: The median is the middle value of a dataset when the observations are arranged in ascending or descending order.

The median divides the ordered data into two equal halves.

Example: For 10, 20, 30, 40, 50, the median is 30. For 10, 20, 30, 40, the median is (20 + 30) / 2 = 25.

4. Mode

Definition: The mode is the value or category that occurs most frequently in a dataset.

A dataset may have one mode, more than one mode, or no mode.

Example: For 10, 20, 20, 20, 30, 40, the mode is 20 because it occurs most often.

5. Measures of Central Tendency

Definition: Measures of central tendency are statistical measures that identify a central or typical value within a dataset.

The three major measures of central tendency are the mean, median, and mode.

6. Range

Definition: The range is a measure of dispersion calculated as the difference between the maximum and minimum values in a dataset.

It gives a simple indication of spread, but it only uses the two extreme values.

Formula: Range = Maximum - Minimum

Example: For 20, 30, 40, 50, 80: Range = 80 - 20 = 60.

7. Variance

Definition: Variance is a measure of dispersion that describes the average squared deviation of observations from their mean.

In simple terms, variance tells us how spread out the observations are around the mean.

Interpretation: Low variance means values are relatively close to the mean; high variance means values are more widely spread.

8. Standard Deviation

Definition: Standard deviation is a measure of dispersion that indicates how much observations typically vary from the mean.

It is the square root of variance and is expressed in the same units as the original data.

Formula: Standard Deviation = square root of Variance

Interpretation: A small standard deviation means values are close to the mean; a large standard deviation means they are more spread out.

9. Measures of Dispersion

Definition: Measures of dispersion describe how spread out or variable the observations in a dataset are.

Common measures of dispersion include range, variance, and standard deviation.

10. Correlation

Definition: Correlation is a statistical measure that describes the strength and direction of the relationship between two variables.

A positive correlation means the variables tend to move in the same direction, while a negative correlation means they tend to move in opposite directions.

Range: The correlation coefficient r ranges from -1 to +1.

Interpretation: r near +1: strong positive linear relationship; r near -1: strong negative linear relationship; r near 0: weak or no linear relationship.

Important: Correlation does not necessarily imply causation.

11. Probability

Definition: Probability is a numerical measure of the likelihood that an event will occur.

Probability ranges from 0 to 1, where 0 means impossible and 1 means certain.

Formula: $0 <= P(A) <= 1$

Example: For a fair coin, P(Heads) = 1/2 = 0.5 = 50%.

12. Event

Definition: An event is a specific outcome or collection of outcomes from a random experiment.

For example, when rolling a die, the event “rolling an even number” contains the outcomes 2, 4, and 6.

13. Conditional Probability

Definition: Conditional probability is the probability of an event occurring given that another event has already occurred or is known to be true.

It is written as P(A | B), which is read as “the probability of A given B.”

Formula: P(A | B) = P(A and B) / P(B)

Example: If 60 students are female and 45 of them passed, then P(Pass | Female) = 45/60 = 0.75 = 75%.

14. Distribution

Definition: A distribution describes how the values of a variable are spread or arranged across their possible values.

It shows which values are common, uncommon, concentrated, or extreme.

15. Normal Distribution

Definition: A normal distribution is a continuous, symmetric, bell-shaped probability distribution in which observations are concentrated around the mean and become less frequent as they move farther from the mean.

In an ideal normal distribution, the mean, median, and mode are equal.

Key properties: Symmetric, bell-shaped, centered at the mean, and observations become less frequent farther from the center.

16. The 68-95-99.7 Rule

Definition: In an approximately normal distribution, about 68% of observations fall within one standard deviation of the mean, about 95% within two standard deviations, and about 99.7% within three standard deviations.

This rule helps describe how values are distributed around the center of a normal distribution.

Example: If mean height = 170 cm and standard deviation = 5 cm, about 68% of values in a normal distribution lie roughly between 165 cm and 175 cm.

17. Population

Definition: A population is the entire group of individuals, objects, or observations that a statistical study is interested in.

For example, all university students in Nepal could be a population for a study about student height.

18. Sample

Definition: A sample is a smaller subset selected from a population for observation and analysis.

Researchers use samples when studying the entire population would be difficult, expensive, or time-consuming.

19. Sampling Distribution

Definition: A sampling distribution is the probability distribution of a statistic, such as the sample mean, obtained from repeated samples of the same size from a population.

It describes how that statistic varies from sample to sample.

Example: Take many samples of the same size, calculate the mean of each sample, and examine the distribution of those means.

20. Central Limit Theorem

Definition: The Central Limit Theorem states that the sampling distribution of the sample mean tends to approach a normal distribution as the sample size becomes sufficiently large, provided appropriate conditions are satisfied, regardless of the exact shape of the original population distribution.

The theorem concerns the distribution of sample means, not the original population itself.

Intuition: Population -> take a random sample -> calculate the sample mean -> repeat many times -> the distribution of sample means tends to look approximately normal.

Important: The CLT does not say that the original population becomes normal. It says that the distribution of sample means tends toward normality under suitable conditions.

Quick Revision Table

Term	Definition
Descriptive Statistics	Methods used to organize, summarize, and describe data.
Mean	Sum of all values divided by the number of observations.
Median	Middle value of an ordered dataset.
Mode	Most frequently occurring value.
Range	Difference between maximum and minimum values.

Variance	Average squared distance of observations from their mean.
Standard Deviation	Typical amount by which observations differ from the mean.
Correlation	Strength and direction of the relationship between two variables.
Probability	Numerical measure of how likely an event is to occur.
Conditional Probability	Probability of one event given that another event has occurred.
Distribution	How observations are spread across possible values.
Normal Distribution	Symmetric, bell-shaped distribution centered around the mean.
Population	Entire group being studied.
Sample	Subset selected from a population.
Sampling Distribution	Distribution of a statistic obtained from repeated samples.
Central Limit Theorem	Sample means tend toward a normal distribution as sample size increases under appropriate conditions.

Key Distinctions to Remember

- Mean, median, and mode describe the center of data.
- Range, variance, and standard deviation describe spread.
- Variance is expressed in squared units; standard deviation is in the original units.
- Correlation describes association, not proof of causation.
- Conditional probability means the probability of one event given information about another event.
- The normal distribution is a shape of a distribution; not every dataset is normal.
- The Central Limit Theorem describes the distribution of sample means, not the original data.